Benchmark paper

Anonymized Authors

Anonymized Affiliations
email@anonymized.com

Abstract. Diffusion MRI is widely used to explore brain microstructure, and numerous machine learning methods have been proposed to extract predictive information. Yet progress remains difficult to assess: studies rely on single datasets, heterogeneous pipelines, limited evaluation, and often lack accessible code, preventing meaningful comparison and reproducibility.
We introduce a modular and transparent benchmarking framework for machine learning in dMRI microstructure. The benchmark unifies multiple representative datasets within a standardized end-to-end pipeline and enables systematic comparison across tissue types, feature representations and models.
It reveals differences between white and gray matter analyses, quantify the impact of diffusion microstructure feature choices, and show that fast linear baselines can rival more complex deep learning methods. Standardized benchmarking is essential to move the field beyond proxy tasks and toward robust methods with real clinical relevance.

Keywords: Benchmark · Brain · diffusion MRI.

1 Introduction (Reuben)

First paragraph:

- Context: Diffusion MRI provides quantitative microstructural biomarkers
- Increasing use of ML for prediction (age, diagnosis, cognition, etc.)
- Lack of standardized benchmark

Second paragraph: literature review:

- Literature review of techniques: Variability of the proposed approach (ML-based, deep-learning based etc)
- Problems: High variability in reported performance across studies + Limited reproducibility

Third paragraph: reasons for these problems

- Reason 1: Heterogeneous preprocessing pipelines
- Reason 2: Inconsistent feature representations (voxel-wise vs ROI-based), not analysis cross microstructure features

- Reason 3: Evaluation protocols not directly comparable + results on different datasets
- Reason 4: Missing code
- Our solution: "To address these challenges, we propose: 1) an open-source and standardized benchmark; 2) of various machine learning techniques in dMRI; 3) on several tasks (classification, regression) 4) encompassing N datasets

Fourth paragraph: your contributions

- Standardized preprocessing pipeline for diffusion MRI
- Implemented various representation of the microstructural input data
- Unified feature extraction strategy
- Controlled evaluation protocol (train/test splits)
- Comparison of multiple feature representations
- Integration of both classical and DL predictors
- Public release of code and benchmark configuration
- End-to-end analysis

2 Benchmark (Reuben)

Start with a general introduction of this section that introduce each component of your benchmark + add a figure that presents an overview.

2.1 Datasets

- Introduction sentence about why you picked these datasets
- Introduce Number of subjects
- Age range / demographics
- Scanner details
- Acquisition protocol or summary table with acquisition parameters
- Prediction targets (classification/regression)

If space (unlikely): * PLOTS: Demographic distribution (age histograms) | Table with demographics summary (gender classes, age range...)

2.2 Unified diffusion data preprocessing

Data preprocessing Following...., we propose to

- Denoising
- Motion and eddy correction
- Spatial normalization
- Masking strategy

Microstructural Representation Explain here the different forms of input representation. The word "feature representation" is ambigiuous because neural network also extract features in a data-driven way. That's why I would rather use the term like "Microstructural Representation". Explain why differences between white and gray matter. Give references here about why there are differences.

Gray Matter Representation

- Voxel-wise features
- High-dimensional representation (64k features)
- Mask-based extraction

Explain that voxel-wise features preserve spatial resolution but increase dimensionality and risk of overfitting.

White Matter Representation

- Tract-based aggregation
- Mean values per tract (48 features)
- Reduced dimensionality
- Anatomical interpretability

Justify tract-averaged representation: Noise reduction, improved stability, lower dimensional feature space, anatomically meaningful structure.

Microstructural normalization

- Binary targets (encoded 0/1), Continuous targets (standardized during training)
- Standardization fitted on training set only
- Applied to validation/test sets
- Inverse transform for regression outputs (for visualization)

2.3 Prediction models

- Dummy baseline (e.g., majority class for classification, mean predictor for regression)
- Classical ML pipeline: Linear models (e.g., logistic regression, linear regression), Random Forests, Support Vector Machines (SVM) + their PCA versions
- Deep Learning models (e.g., dinov2, medicalnet)
- Unified hyperparameter search strategy (e.g., grid search or random search with cross-validation on the training set)

2.4 Benchmarks tasks and evaluation metrics

Task selection rationale: Clinical relevance, data availability, and established benchmarks in the literature.

Classification

- Task description: Binary diagnosis prediction (e.g., ASD vs. TD)
- Evaluation Metrics: Accuracy, AUC, F1-score

Regression

- Task description: Age prediction (brain age estimation)
- Evaluation Metrics: R^2, MAE

2.5 Evaluation protocols

- Data Splitting Strategy (Fixed train/test splits)
- Cross-validation within training set. Stratification
- No information leakage
- Mean ± std across folds
- Statistical comparison tests (e.g., paired t-tests, Wilcoxon signed-rank tests) to assess significance of performance differences between models

3 Results (Reuben)

Introduction sentence

Choice of prediction models Table: For all tasks, choose one brain structure and one microstructural representation (choose widely this one) and report results in a table + comment

Choice of the brain structure For one regression task and one classification task, show the impact of the brain structure with one miscrostructural representation. You can discard models that underperformed in the first model section.

Choice of the microstructural representation For one task, show the impact of the microstructural representation for each brain structure. You can discard models that underperformed in the first model section.

Training size impact I think we will not have space for this, but here it would be for one regression task and one classification task, one structure, one microstrucral representation, impact of the model, here try to have variability (PCA, non-PCA, deep) for example.

Methodological insights

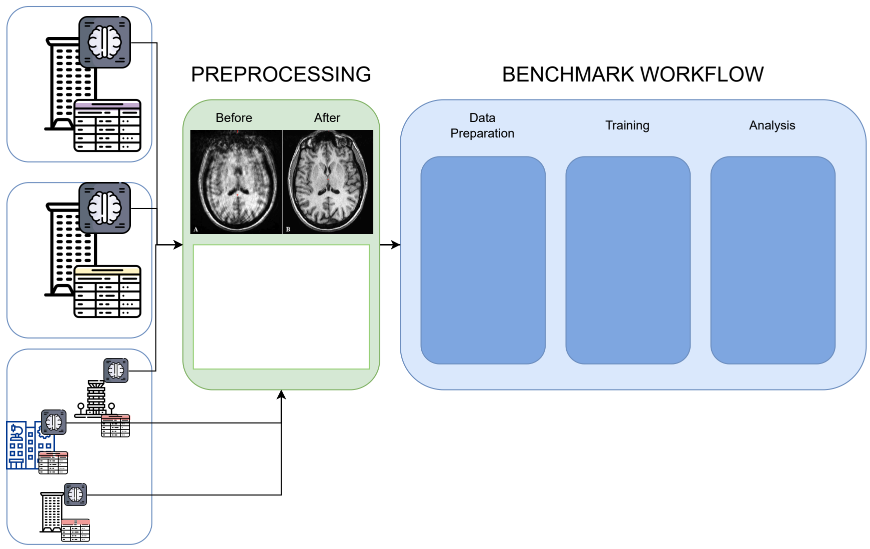
- DL vs ML based
- Importance of dimensionality control
- Effect of tract-based aggregation
- Risk of overfitting in mesh representations without PCA bcs of the high number of vertices representations

4 Conclusion (Reuben)

- Introduced standardized dMRI benchmark for age prediction in healthy individuals using three large datasets (HCP, CamCAN, ABIDE)
- Compared feature representations (MD, SH, MK, RTOP) and models (dummy classifier, linear, pca_linear, forest, pca_forest, svm, pca_svm)
- Provided reproducible end-to-end framework and enables fair model comparison. Reduces methodological heterogeneity
- A step toward methodological unification/code availability

5 Introduction

5.1 Scientific and Clinical Motivation



- Diffusion MRI provides quantitative microstructural biomarkers
- Increasing use of ML for prediction (age, diagnosis, cognition, etc.)
- Lack of standardization in preprocessing and feature representation
- High variability in reported performance across studies

Table idea

Study	Preprocessing	Standardized?	Feature Type	Public Code?
Study A	Pipeline A	No	Voxel-wise	No
Study B	Pipeline B	No	ROI-based	Yes

5.2 Gap in the Field

- No standardized benchmark for diffusion MRI prediction
- Heterogeneous preprocessing pipelines
- Inconsistent feature representations (voxel-wise vs ROI-based), not analysis cross microstructure features
- Evaluation protocols not directly comparable
- Limited reproducibility and transparency, missing code
- Lack of results of the same experiments on other datasets

5.3 Contributions

- Standardized preprocessing pipeline for diffusion MRI
- Unified feature extraction strategy
- Controlled evaluation protocol (train/test splits)
- Comparison of multiple feature representations
- Public release of code and benchmark configuration
- End-to-end analysis
- Integration of both classical and DL models

6 Benchmark

Start with a general introduction of this section that introduce each component of your benchmark + add a figure that presents an overview.

Our benchmark agregates datasets with each dataset preprocessed with the same pipeline, with the exception of HCP which has its own preprocessing pipeline. We perform the feature extraction/data preparation and apply rigorously the analysis and cross validation that we are going to explain.

6.1 Datasets

To ensure reproducibility, we evaluate our benchmark on three publicly available datasets covering both healthy and clinical populations, offering diverse demographics, scanner types, and acquisition protocols. For healthy cohorts, we utilize the Human Connectome Project (HCP) Young Adult dataset and the Cambridge Centre for Ageing and Neuroscience (Cam-CAN) dataset, both targeting gender and age prediction (classification and regression, respectively). After filtering, the HCP cohort comprises 1,036 subjects (474 males, 562 females; ages 22-37). HCP data were acquired on a customized Siemens 3T Connectome scanner using a multi-shell Spin-Echo EPI sequence (TR/TE = 5520/89.5 ms, 1.25 mm isotropic, multiband factor 3; b-values = 1000, 2000, 3000 s/mm^2, 90 directions/shell). The Cam-CAN cohort includes 629 subjects (316 males, 313 females; ages 18-88), scanned on a Siemens 3T TIM Trio using a 2D twice-refocused SE EPI sequence (TR/TE = 9100/104 ms, 2.0 mm isotropic; b-values = 1000, 2000 s/mm^2, 30 directions each).

For clinical evaluation, we use the Autism Brain Imaging Data Exchange II (ABIDE II) dataset, chosen over ABIDE I due to the availability of diffusion MRI. Targeting Autism Spectrum Disorder (ASD) diagnosis (classification), the filtered ABIDE II cohort consists of 218 subjects (186 males, 32 females; ages 5-64), **including 119 typically developing individuals and 99 with ASD (NEEDS TO VERIFY)**. As a multi-site dataset, ABIDE II utilizes various 3T scanners (e.g., Siemens, GE) with site-specific single-shell diffusion protocols, providing a realistic scenario for evaluating model robustness to scanner heterogeneity.

If space (unlikely): * PLOTS: Demographic distribution (age histograms) | Table with demographics summary (gender classes, age range...)

6.2 Preprocessing and Feature Extraction

Preprocessing (Himanshu)

- Denoising
- Motion and eddy correction
- Spatial normalization
- Masking strategy

Microstructural Representation

Gray Matter Representation Gray matter microstructure was represented on the subject-specific cortical surface reconstructed from **T1-weighted images using the Desikan–Killiany atlas (aparc) (REVIEW)**. For each subject, cortical meshes (white/pial surfaces) were generated and diffusion-derived scalar maps (e.g., MD, SH-based metrics) were computed in native diffusion space and rigidly aligned to the anatomical space. Voxelwise diffusion measures were then projected onto the individual cortical surface using ribbon-constrained sampling, preserving vertex-wise resolution (64k vertices per hemisphere). This subject-specific surface representation was chosen to respect cortical geometry, reduce partial-volume effects, and maintain high spatial specificity while enabling anatomically consistent regional correspondence via the atlas labels across subjects.

White Matter Representation White matter microstructure was represented within a standardized tract space using the Tract-Based Spatial Statistics (TBSS) framework. Diffusion-derived scalar maps were nonlinearly registered to standard space, projected onto the group-wise white matter skeleton to minimize partial volume effects and residual misalignment, and summarized using the JHU White-Matter Tractography Atlas. Mean microstructural values were extracted from 48 major bundles defined in the atlas, yielding a compact tract-level representation per metric. We intentionally adopted this skeleton-based, atlas-constrained strategy rather than subject-specific tractography to enforce anatomical correspondence across subjects, reduce variability introduced by tracking parameters, and enhance robustness in a multi-dataset benchmarking setting. Although this representation sacrifices voxel-level granularity relative to the cortical surface analysis, it provides reproducible, homologous bundle definitions across datasets and mitigates registration and tract delineation inconsistencies—thereby strengthening cross-cohort comparability and methodological stability.

6.3 Prediction

- Feature normalization
  - Binary targets (encoded 0/1), Continuous targets (standardized during training)
  - Standardization fitted on training set only

  - Applied to validation/test sets
  - Inverse transform for regression outputs (for visualization)
- Task definitions and labels and reasons for task selection
  - Classification: Binary diagnosis prediction (e.g., ASD vs. TD)
  - Regression: Age prediction (brain age estimation)
  - Task selection rationale: Clinical relevance, data availability, and established benchmarks in the literature
- Evaluation Metrics
  - Classification: Accuracy, AUC, F1-score
  - Regression: R^2, MAE
  - Mean $\pm$ std across folds
  - Statistical comparison tests (e.g., paired t-tests, Wilcoxon signed-rank tests) to assess significance of performance differences between models
- Models Used
  - Dummy baseline (e.g., majority class for classification, mean predictor for regression)
  - Classical ML pipeline: Linear models (e.g., logistic regression, linear regression), Random Forests, Support Vector Machines (SVM) + their PCA versions
  - Deep Learning models (e.g., dinov2, medicalnet)
  - Unified hyperparameter search strategy (e.g., grid search or random search with cross-validation on the training set)

6.4 Validation, Evaluation and Visualizations

- Data Splitting Strategy (Fixed train/test splits)
- Cross-validation within training set. Stratification
- No information leakage
- Explanation of plots and statistics

7 Results

7.1 Within-Dataset Performance

- Model/Feature/Tissue comparison
- Mean $\pm$ std of best run / model
- Statistical testing
- PLOT: Bar plots per dataset | Camcan-HCP comparison(?

7.2 Feature Representation Comparison

- Gray vs white matter
- Microstructure metrics comparison
- PLOT: Tissue performance according to microostructure metric for task purposes | Microstructure features

7.3 "Handmade" vs DL embeddings

7.4 Training Size Analysis

* PLOT: Learning Curve

7.5 Key Findings

- Which models perform best? Which feature representations perform best? Tissue predominance in task prediction?
- Are simpler models sufficient? Are DL models necessary for predicting demographics?
- Stability across folds?

7.6 Methodological Insights

- Importance of dimensionality control
- Effect of tract-based aggregation
- Risk of overfitting in mesh representations without PCA bcs of the high number of vertices representations

8 Conclusion

- Introduced standardized dMRI benchmark for age prediction in healthy individuals using three large datasets (HCP, CamCAN, ABIDE)
- Compared feature representations (MD, SH, MK, RTOP) and models (dummy classifier, linear, pca_linear, forest, pca_forest, svm, pca_svm)
- Provided reproducible end-to-end framework and enables fair model comparison. Reduces methodological heterogeneity
- A step toward methodological unification/code availability

8.1 Limitations/Further steps

- No multi-site harmonization (explicitly as scope limitation). Would training in one dataset and predicting in another dataset have the same effect as predicting in different folds of the same dataset? Are the predictions very far?
- Limited tasks / ranges in prediction (e.g. HCP age prediction)
- Limited preprocessing / including other preprocessing parameters or corrections

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], and a homepage [5]. Multiple citations are grouped [1–3], [1, 3–5].

Acknowledgments. A bold run-in heading in small font size at the end of the paper is used for general acknowledgments, for example: This study was funded by X (grant number Y).

Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments[1], for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

References

1. Author, F.: Article title. Journal **2**(5), 99–110 (2016)
2. Author, F., Author, S.: Title of a proceedings paper. In: Editor, F., Editor, S. (eds.) CONFERENCE 2016, LNCS, vol. 9999, pp. 1–13. Springer, Heidelberg (2016). https://doi.org/10.10007/1234567890
3. Author, F., Author, S., Author, T.: Book title. 2nd edn. Publisher, Location (1999)
4. Author, A.-B.: Contribution title. In: 9th International Proceedings on Proceedings, pp. 1–2. Publisher, Location (2010)
5. LNCS Homepage, http://www.springer.com/lncs, last accessed 2023/10/25

[1] If EquinOCS, our proceedings submission system, is used, then the disclaimer can be provided directly in the system.